Physics 9702 Paper 4 · Topic 17

Oscillations

The seven hardest A Level Structured Questions on this topic from the 2020 to 2026 papers, hardest first, followed by their mark schemes.

---

1 9702/42 February/March 2022, Question 3 (10 marks) page 2

Examiner report: A common misconception was an incorrect reference to Newton's Third Law, and an explanation of upwards resultant force as a necessarily equal and opposite response to the downwards push. (c) (i) There were many good definitions of simple harmonic motion, but very few candidates related this to the stated force equation.

2 9702/41 May/June 2021, Question 3 (8 marks) page 5

Examiner report: A significant number of candidates showed a misconception that the amplitude was constant at all frequencies.

3 9702/41 May/June 2022, Question 4 (8 marks) page 7

Examiner report: (c) This explanation proved challenging. Only the strongest candidates managed to gain full credit because they realised that ω would decrease leading to a consequent increase in amplitude.

4 9702/41 October/November 2021, Question 4 (11 marks) page 9

Examiner report: (c) This sketch graph proved to be challenging. Only the strongest candidates drew a closed loop based about (20, 0).

5 9702/42 October/November 2021, Question 4 (7 marks) page 11

Examiner report: (ii) This was a more challenging question, and only the strongest candidates realised that the squared nature of the variation of potential energy with displacement (or kinetic energy with velocity) means that the displacement where these quantities have half their maximum value is more than half of the amplitude.

6 9702/42 May/June 2021, Question 3 (9 marks) page 13

Examiner report: (c) Candidates found this question difficult.

7 9702/44 October/November 2025, Question 4 (10 marks) page 15

Examiner report: Only the strongest candidates mentioned that kinetic energy + potential energy is constant.

Questions were chosen using the Cambridge examiner reports (Principal Examiner Report for Teachers): parts the reports describe as challenging, poorly answered, or answered fully only by the strongest candidates. May/June 2020 and May/June 2026 have no report, so questions from those series were judged directly. Variant 43 repeats variant 41 in every series, so only 41 is cited.

3 A small wooden block (cuboid) of mass m floats in water, as shown in Fig. 3.1.

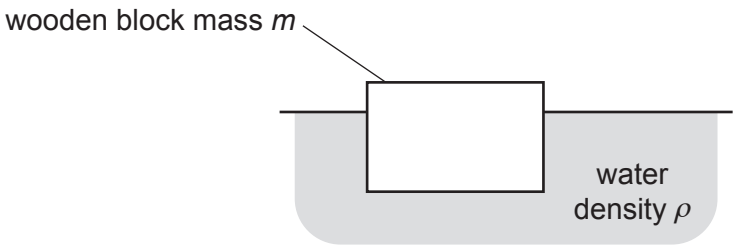


Fig. 3.1

The top face of the block is horizontal and has area A. The density of the water is ρ.

(a) State the names of the two forces acting on the block when it is stationary.

..... [1]

(b) The block is now displaced downwards as shown in Fig. 3.2 so that the surface of the water is higher up the block.

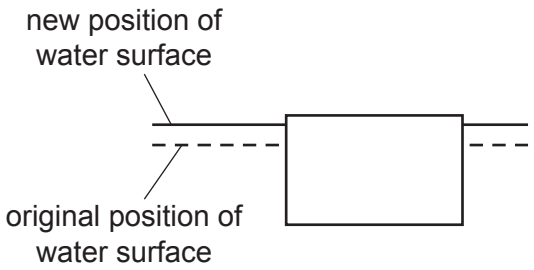


Fig. 3.2

State and explain the direction of the resultant force acting on the wooden block in this position.

.....

..... [1]

(c) The block in **(b)** is now released so that it oscillates vertically.

The resultant force F acting on the block is given by

$$F = -Ag\rho x$$

where g is the gravitational field strength and x is the vertical displacement of the block from the equilibrium position.

(i) Explain why the oscillations of the block are simple harmonic.

.....

.....

..... [2]

(ii) Show that the angular frequency ω of the oscillations is given by

$$\omega = \sqrt{\frac{A\rho g}{m}}.$$

[2]

(d) The block is now placed in a liquid with a greater density. The block is displaced and released so that it oscillates vertically. The variation with displacement x of the acceleration a of the block is measured for the first half oscillation, as shown in Fig. 3.3.

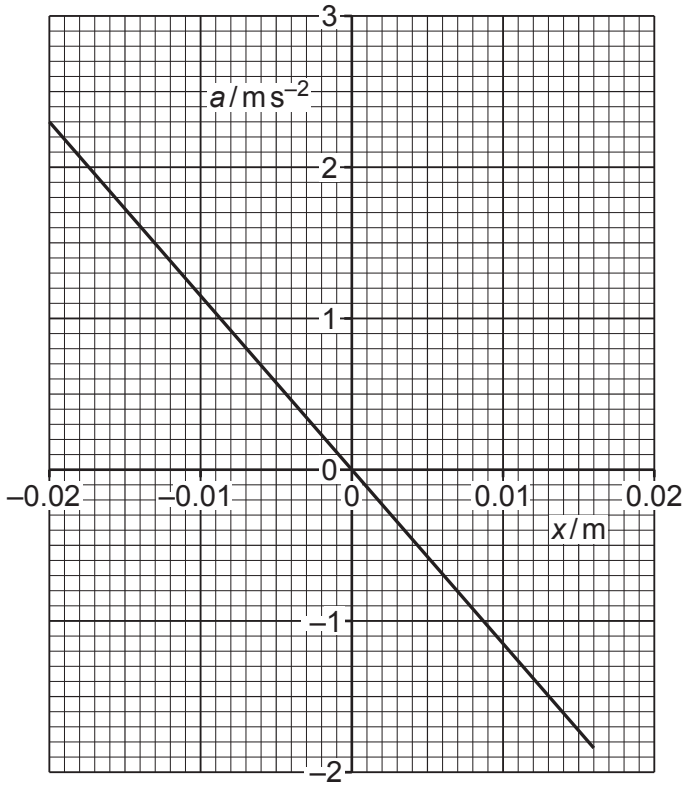


Fig. 3.3

(i) Explain why the maximum negative displacement of the block is not equal to its maximum positive displacement.

.....

.....

..... [1]

(ii) The mass of the block is 0.57 kg.

Use Fig. 3.3 to determine the decrease ΔE in energy of the oscillation for the first half oscillation.

E = ... J [3]

[Total: 10]

3 **(a)** State what is meant by *simple harmonic motion*.

.....

.....

..... [2]

(b) A trolley of mass *m* is held on a horizontal surface by means of two springs. One spring is attached to a fixed point P. The other spring is connected to an oscillator, as shown in Fig. 3.1.

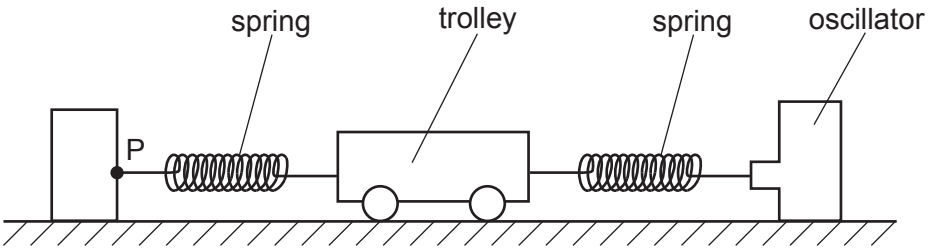


Fig. 3.1

The springs, each having spring constant *k* of 130 N m^{-1}, are always extended.

The oscillator is switched off. The trolley is displaced along the line of the springs and then released. The resulting oscillations of the trolley are simple harmonic.

The acceleration *a* of the trolley is given by the expression

$$a = -\left(\frac{2k}{m}\right)x$$

where *x* is the displacement of the trolley from its equilibrium position.

The mass of the trolley is 840 g.

Calculate the frequency *f* of oscillation of the trolley.

f = ... Hz [3]

(c) The oscillator in **(b)** is switched on. The frequency of oscillation of the oscillator is varied, keeping its amplitude of oscillation constant.

The amplitude of oscillation of the trolley is seen to vary. The amplitude is a maximum at the frequency calculated in **(b)**.

(i) State the name of the effect giving rise to this maximum.

..... [1]

(ii) At any given frequency, the amplitude of oscillation of the trolley is constant.

Explain how this indicates that there are resistive forces opposing the motion of the trolley.

.....

.....

..... [2]

[Total: 8]

4 A pendulum consists of a bob (small metal sphere) attached to the end of a piece of string. The other end of the string is attached to a fixed point. The bob oscillates with small oscillations about its equilibrium position, as shown in Fig. 4.1.

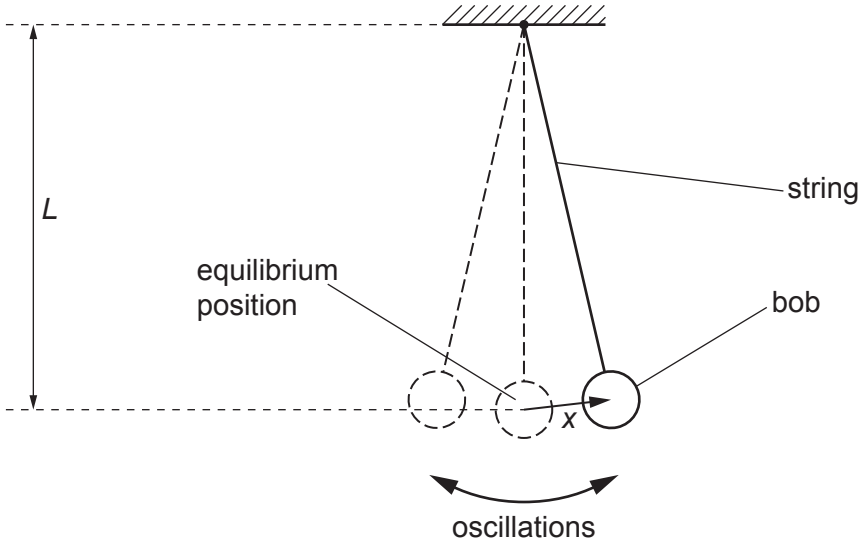


Fig. 4.1 (not to scale)

The length *L* of the pendulum, measured from the fixed point to the centre of the bob, is 1.24 m.

The acceleration *a* of the bob varies with its displacement *x* from the equilibrium position as shown in Fig. 4.2.

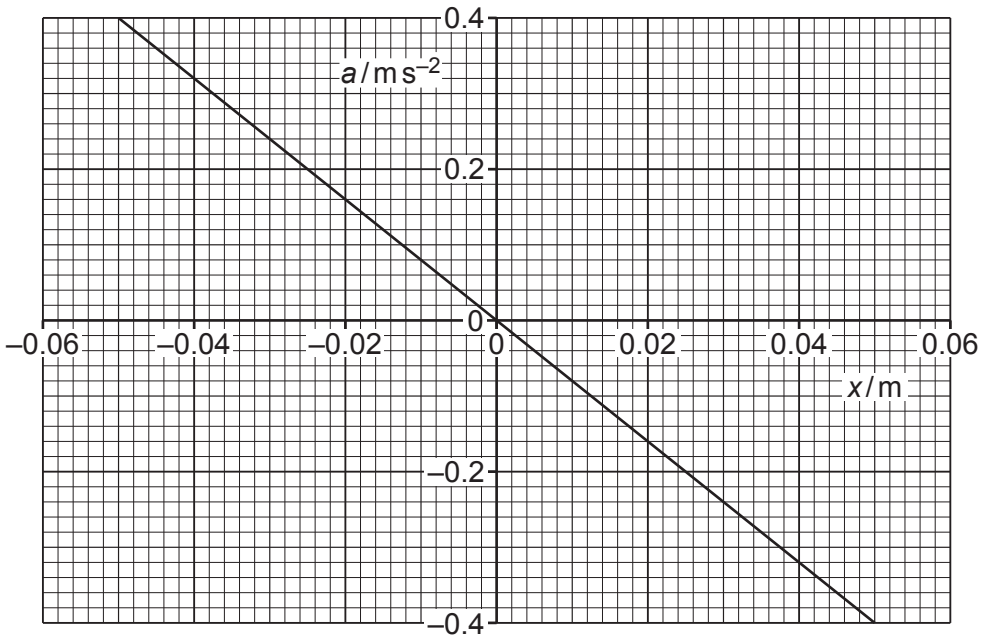


Fig. 4.2

(a) State how Fig. 4.2 shows that the motion of the pendulum is simple harmonic.

.....

.....

..... [2]

(b) **(i)** Use Fig. 4.2 to determine the angular frequency ω of the oscillations.

ω = ... rad s^{-1} [2]

(ii) The angular frequency ω is related to the length L of the pendulum by

$$\omega = \sqrt{\frac{k}{L}}$$

where k is a constant.

Use your answer in **(b)(i)** to determine k. Give a unit with your answer.

k = ... unit ..... [2]

(c) While the pendulum is oscillating, the length of the string is increased in such a way that the total energy of the oscillations remains constant.

Suggest and explain the qualitative effect of this change on the amplitude of the oscillations.

.....

.....

..... [2]

[Total: 8]

4 A trolley on a track is attached by springs to fixed blocks X and Y, as shown in Fig. 4.1. The track contains many small holes through which air is blown vertically upwards. This results in the trolley resting on a cushion of air rather than being in direct contact with the track.

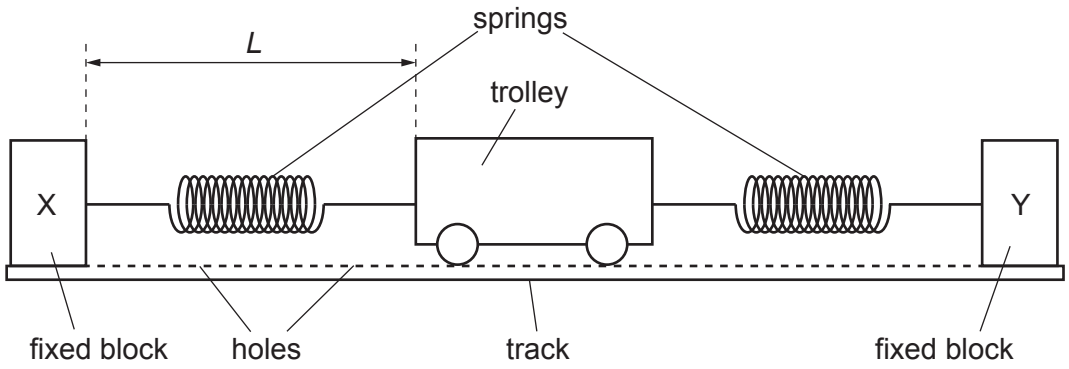


Fig. 4.1

The trolley is pulled to one side of its equilibrium position and then released so that it oscillates initially with simple harmonic motion. After a short time, the air blower is switched off. The variation with time *t* of the distance *L* of the trolley from block X is shown in Fig. 4.2.

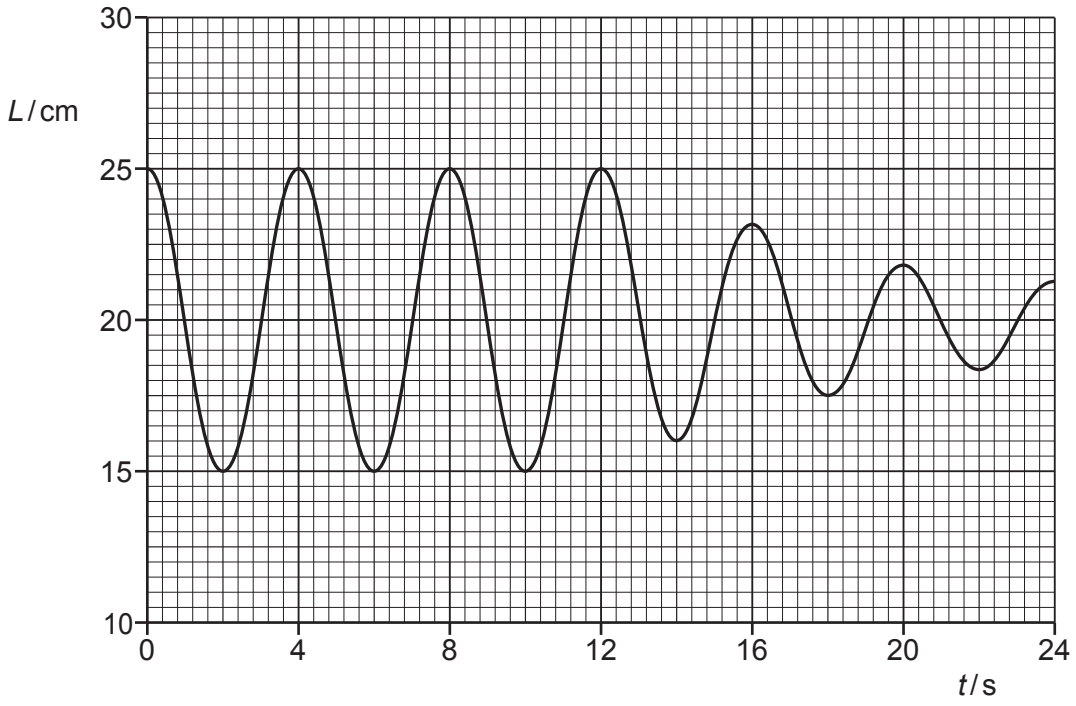


Fig. 4.2

(a) Use Fig. 4.2 to determine:

(i) the initial amplitude of the oscillations

amplitude = ... cm [1]

(ii) the angular frequency ω of the oscillations

ω = ... rad s^{-1} [2]

(iii) the maximum speed v_0, in cm s^{-1}, of the oscillating trolley.

v_0 = ... cm s^{-1} [2]

(b) Apart from the quantities in **(a)**, describe what may be deduced from Fig. 4.2 about the motion of the trolley between time t = 0 and time t = 24 s. No calculations are required.

.....

.....

.....

.....

..... [3]

(c) On Fig. 4.3, sketch the variation with L of the velocity v of the trolley for its first complete oscillation.

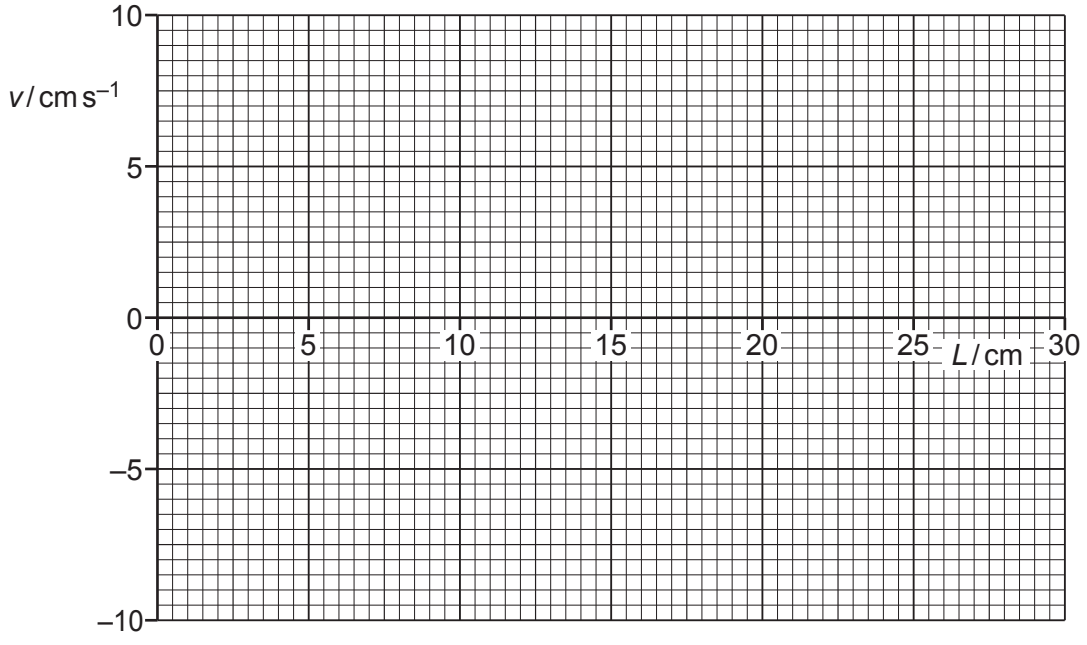


Fig. 4.3

[3]

[Total: 11]

4 A trolley on a smooth surface is attached by springs to fixed blocks as shown in Fig. 4.1.

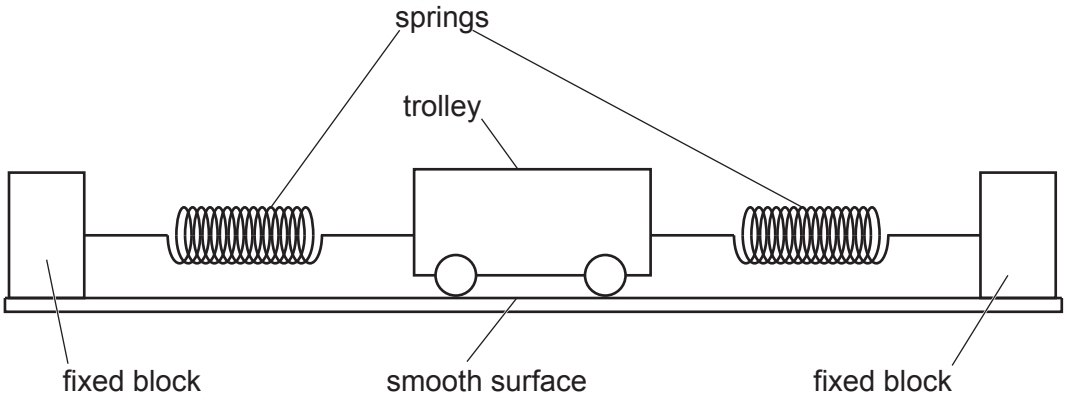


Fig. 4.1

The trolley oscillates horizontally about its equilibrium position with an amplitude of 12 cm. Fig. 4.2 shows the variation of the acceleration a of the trolley with displacement x from its equilibrium position. Friction between the trolley and the surface can be assumed to be negligible.

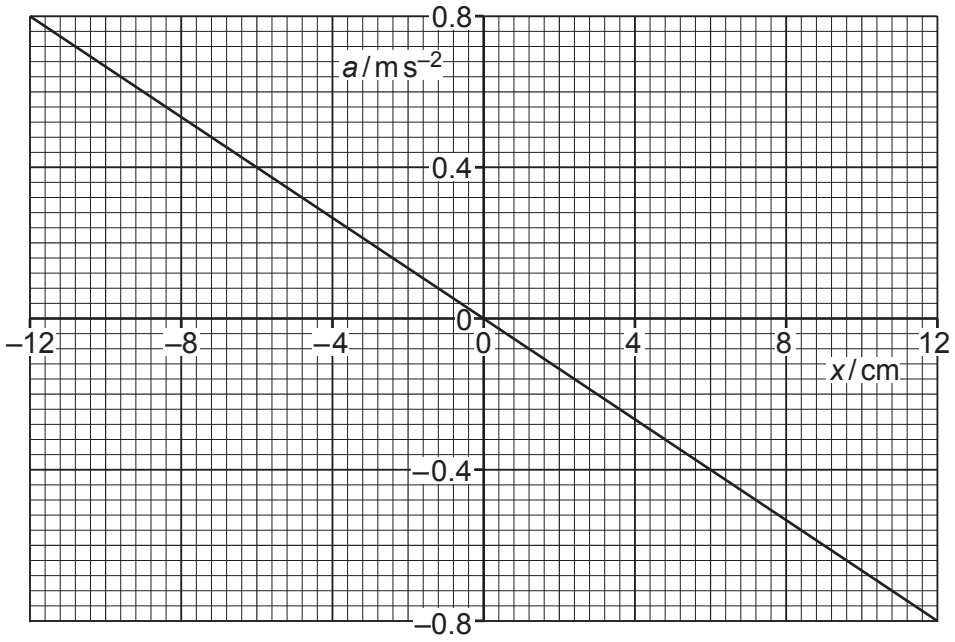


Fig. 4.2

(a) Describe the features of the line in Fig. 4.2 that demonstrate that the motion of the trolley is simple harmonic.

.....

.....

..... [2]

(b) Use Fig. 4.2 to determine the period T of the oscillations of the trolley.

T = ... s [3]

(c) **(i)** On the line of the graph of Fig. 4.2, label with the letter P one point where the kinetic energy of the trolley is zero. [1]

(ii) On the line of the graph of Fig. 4.2, label with the letter Q an approximate position of one point where the kinetic energy of the trolley is equal to the potential energy stored in the springs. [1]

[Total: 7]

3 A U-shaped tube contains some liquid. The liquid column in each half of the tube has length *L*, as shown in Fig. 3.1.

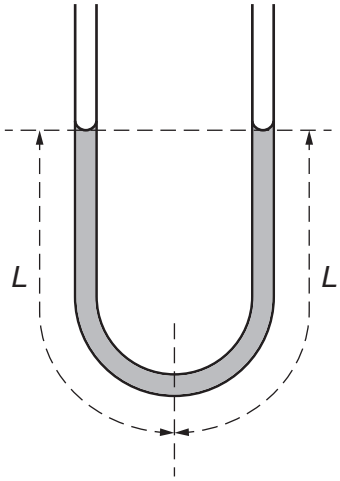


Fig. 3.1

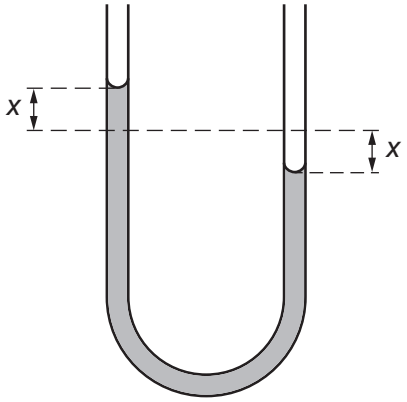


Fig. 3.2

The liquid columns are displaced vertically. The liquid then oscillates in the tube. The liquid levels are displaced from the equilibrium positions as shown in Fig. 3.2.

The acceleration *a* of the liquid in the tube is related to the displacement *x* by the expression

$$a = -\left(\frac{g}{L}\right)x$$

where *g* is the acceleration of free fall.

(a) Explain how the expression shows that the liquid in the tube is undergoing simple harmonic motion.

.....

.....

.....

.....

..... [3]

(b) The length *L* of each liquid column is 18 cm.

Determine the period *T* of the oscillations.

T = ... s [3]

(c) The oscillations of the liquid in the tube are damped.
In any one complete cycle of the oscillations, the amplitude decreases by 6.0% of its value at the beginning of the oscillation.

Determine the ratio

$$\frac{\text{energy of oscillations after 3 cycles}}{\text{initial energy of oscillations}}.$$

ratio = ... [3]

[Total: 9]

4 **(a)** State what is meant by the frequency of the oscillations of an oscillating object.

.....

..... [1]

(b) An object is oscillating.

Fig. 4.1 shows the variation of the acceleration a of the object with its displacement x from the equilibrium position.

Fig. 4.2 shows the variation of the kinetic energy E_K of the object with time t.

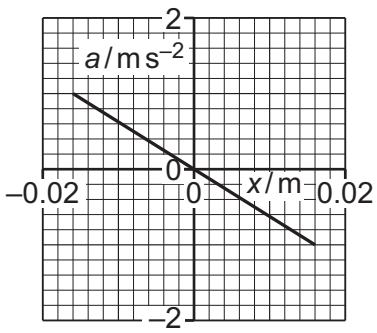


Fig. 4.1

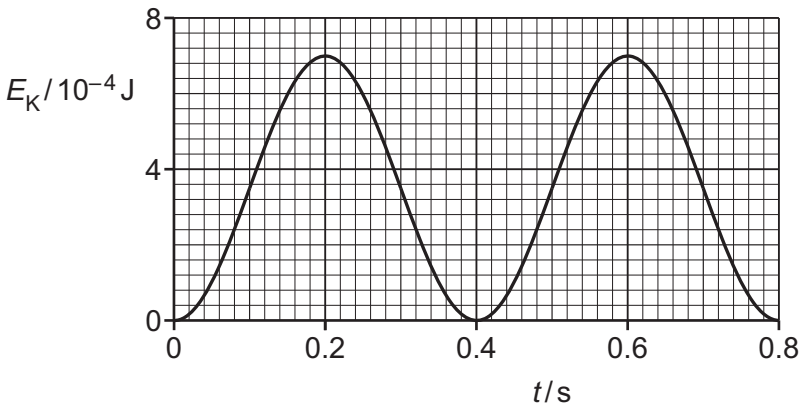


Fig. 4.2

(i) Explain how Fig. 4.2 shows that the period of the oscillations is 0.80 s.

.....

.....

..... [1]

(ii) Calculate the angular frequency ω of the oscillations.

ω = ... rad s^{-1} [2]

(iii) Apart from the period, frequency and angular frequency of the oscillations, determine **three** other conclusions about the object and its oscillations that may be drawn from Fig. 4.1 and Fig. 4.2. The conclusions may be qualitative or quantitative. Use the space below for any working.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(iv) Describe the interchange between kinetic energy and potential energy during the oscillations. Numerical values are **not** required.

.....

.....

.....

.....

..... [3]

[Total: 10]

Question	Answer	Marks
3(a)	upthrust, weight	**B1**
3(b)	upthrust greater than weight so (resultant force is) upwards	**B1**
3(c)(i)	A, g and ρ all constant so $F \propto x$	**B1**
	minus sign means F and x are in opposite directions	**B1**
3(c)(ii)	$(a = \frac{F}{m}$ so) $a = (-)\frac{Ag\rho x}{m}$	**M1**
	so $\omega^2 = \frac{Ag\rho}{m}$ hence $\omega = \sqrt{\frac{Ag\rho}{m}}$	**A1**
3(d)(i)	damping due to viscous forces	**B1**
3(d)(ii)	$(E =) \frac{1}{2} m\omega^2 x_0^2$	**C1**
	ω^2 = (–) gradient	**C1**
	$(E =) \frac{1}{2} m\omega^2 (x_1^2 - x_2^2)$ $= \frac{1}{2} \times 0.57 \times (\frac{2.3}{0.020})(0.020^2 - 0.016^2)$ $= 4.7 \times 10^{-3}$ J	**A1**

Question	Answer	Marks
3(a)	acceleration (directly) proportional to displacement	**B1**
	acceleration is in opposite direction to displacement	**B1**
3(b)	$\omega^2 = 2k / m$ **and** $\omega = 2\pi f$	**C1**
	$(2\pi f)^2 = (2 \times 130) / 0.84$	**C1**
	$f = 2.8$ Hz	**A1**
3(c)(i)	resonance	**B1**
3(c)(ii)	oscillator supplies energy (continuously)	**B1**
	energy of trolley constant so energy must be dissipated **or** without loss of energy the amplitude would continuously increase	**B1**

Question	Answer	Marks
4(a)	straight line through origin shows that *a* is proportional to *x*	**B1**
	negative gradient shows that *a* is in opposite direction to *x*	**B1**
4(b)(i)	$a_0 = \omega^2 x_0$ **or** $a = -\omega^2 x$ **or** ω^2 = – gradient	**C1**
	$\omega = \sqrt{(0.40 / 0.050)}$ $= 2.8$ rad s^{-1}	**A1**
4(b)(ii)	$k = \omega^2 L$ $= 2.8^2 \times 1.24$	**C1**
	$= 9.7$ m s^{-2}	**A1**
4(c)	(increasing *L* causes) ω to decrease **or** energy (= ½ $m\omega^2 x_0^2$) = ½ mkx_0^2 / L (and *L* increases)	**M1**
	so amplitude increases	**A1**

Question	Answer	Marks
4(a)(i)	5.0 cm	**A1**
4(a)(ii)	$\omega = 2\pi / T$ **or** $\omega = 2\pi f$ **and** $f = 1 / T$	**C1**
	$\omega = 2\pi / 4.0$ $= 1.6$ rad s^{-1}	**A1**
4(a)(iii)	$v_0 = \omega x_0$	**C1**
	$= 1.57 \times 5.0$ $= 7.9$ cm s^{-1}	**A1**
4(b)	• initial pull was to the right • distance from X to trolley (at equilibrium) is 20 cm • period is 4.0 s • initial motion undamped • motion becomes damped at/from 12 s • damping is light • maximum speed at 1 s, 3 s, etc. / stationary at 2 s, 4 s, etc. *Any three points, 1 mark each*	**B3**
4(c)	sketch: closed loop encircling (20, 0)	**B1**
	minimum L shown as 15 cm and maximum L shown as 25 cm	**B1**
	minimum v shown as –7.9 cm s^{-1} and maximum v shown as +7.9 cm s^{-1}	**B1**

Question	Answer	Marks
4(a)	straight line through the origin	**B1**
	negative gradient	**B1**
4(b)	$a = (-)\omega^2 x$ **and** $T = 2\pi / \omega$	**C1**
	e.g. $\omega = \sqrt{(0.80 / 0.12)}$ (*any correct pair of values of a and x*) (= 2.58 rad s^{-1})	**C1**
	$T = 2\pi / 2.58$ = 2.4 s	**A1**
4(c)(i)	Point labelled P at one end of the line	**B1**
4(c)(ii)	Point labelled Q at displacement with magnitude more than half but less than maximum	**B1**

Question	Answer	Marks
3(a)	acceleration in opposite direction to displacement shown by – sign	**B1**
	g / L is constant	**M1**
	(so) acceleration is (directly) proportional to displacement	**A1**
3(b)	$\omega^2 = g / L$	**C1**
	$\omega = 2\pi / T$ **or** $\omega = 2\pi f$ **and** $f = 1 / T$	**C1**
	$(2\pi / T)^2 = 9.81 / 0.18$ $T = 0.85$ s	**A1**
3(c)	energy $\propto x_0^2$	**C1**
	(after 3 cycles,) amplitude $= (0.94)^3 x_0$ $= 0.83 x_0$	**C1**
	ratio final energy / initial energy $= 0.83^2$ $= 0.69$	**A1**

Question	Answer	Marks
4(a)	number of oscillations per unit time	**B1**
4(b)(i)	kinetic energy (of object) reaches maximum / minimum / zero twice in a cycle	**A1**
4(b)(ii)	$\omega = 2\pi / T$ $= 2\pi / 0.80$ **or** $a_0 = \omega^2 x_0$ $\omega = \sqrt{(1.0 / 0.016)}$	**C1**
	$\omega = 7.9$ rad s^{-1}	**A1**
4(b)(iii)	*Any three points from:* • oscillations are simple harmonic • amplitude = 0.016 m • maximum speed = 0.13 m s^{-1} • total energy of oscillations = 7.0×10^{-4} J • mass of object = 0.087 kg • maximum momentum = 0.011 kg m s^{-1} **or** 0.011 N s	**B3**

Question	Answer	Marks
4(b)(iv)	*Any two points from:* • kinetic energy is a maximum at zero displacement **or** kinetic energy is zero at maximum displacement • potential energy is zero at zero displacement **or** potential energy is a maximum at maximum displacement • kinetic energy is maximum when the potential energy is zero **or** potential energy is a maximum when the kinetic energy is zero	**B2**
	kinetic energy + potential energy is constant	**B1**